AP Precalculus: 2026–2027

Unit 3: Trigonometric and Polar Functions Lesson 3.9: Inverse Trigonometric Functions

Primary Objective: Students construct analytical and graphical representations of the inverse sine, cosine, and tangent functions over restricted domains, and interpret output values as angle measures. (2–3 instructional periods, 55 minutes each)

Priority Ideas & Skills

AP Skills

- **1.C** — Construct new functions (inverses)
- **2.B** — Construct equivalent representations

Learning Objective

- **3.9.A** — Construct analytical and graphical representations of inverse sine, cosine, and tangent over restricted domains

Key Understandings (EKs)

- Input/output switch: output of inverse trig = angle measure
- Trig functions are periodic (many-to-one) $\Rightarrow$ must restrict domain to invert
- Restricted domains: $\sin\colon [-\frac{\pi}{2}, \frac{\pi}{2}]$, $\cos\colon [0, \pi]$, $\tan\colon (-\frac{\pi}{2}, \frac{\pi}{2})$

Vocabulary, Concepts, & Theorems

- **Arcsine** ($\arcsin x$ or $\sin^{-1} x$) — inverse of the restricted sine function
- **Arccosine** ($\arccos x$ or $\cos^{-1} x$) — inverse of the restricted cosine function
- **Arctangent** ($\arctan x$ or $\tan^{-1} x$) — inverse of the restricted tangent function
- **Restricted domain** — a subset of the natural domain chosen so the function is one-to-one
- **Principal value** — the unique output angle in the restricted range of an inverse trig function
- **One-to-one function** — each output value corresponds to exactly one input value (passes horizontal line test)

Activate Prior Knowledge & Spiral Review

Warm-Up reviews:

- Evaluating $\sin\theta$, $\cos\theta$, $\tan\theta$ at multiples of $\frac{\pi}{6}$ and $\frac{\pi}{4}$
- Domain and range vocabulary for a function
- Identifying one-to-one functions (horizontal line test) from a table of values

See attached warm-up.

Hook

Display: *"A ramp rises 3 feet over a horizontal run of 5 feet. What angle does the ramp make with the ground?"* Students quickly realize they can compute $\tan\theta = \frac{3}{5}$, but need to *undo* the tangent to find θ. Ask: *"What operation undoes addition? What operation undoes tangent?"* Introduce the idea of an inverse function for trigonometry.

Lesson

Day 1 — Why Restricted Domains?

- Use the table of sin values to show that $\sin\frac{\pi}{6} = \sin\frac{5\pi}{6} = \frac{1}{2}$. Ask: can we define an inverse if two inputs give the same output?
- Introduce the horizontal line test; connect to one-to-one functions from Unit 1.
- Restrict: sine to $[-\frac{\pi}{2}, \frac{\pi}{2}]$, cosine to $[0, \pi]$, tangent to $(-\frac{\pi}{2}, \frac{\pi}{2})$. Ask students why these particular intervals.

Day 2 — Defining & Evaluating Inverse Trig

- Fill in the restricted-domain table in the notes (Section 2).
- Model evaluation: $\arcsin\frac{1}{2} = ?$ — "Which angle in $[-\frac{\pi}{2}, \frac{\pi}{2}]$ has sine equal to $\frac{1}{2}$?" Answer: $\frac{\pi}{6}$.
- Work through the notes evaluation table (Section 3) together.
- Discuss notation: $\sin^{-1}$ means *inverse function*, not $\frac{1}{\sin}$.

Day 3 — Compositions

- When does $\sin(\arcsin x) = x$? (Always, for $x \in [-1, 1]$.)
- When does $\arcsin(\sin x) = x$? (Only for $x \in [-\frac{\pi}{2}, \frac{\pi}{2}]$.)
- Model a composite like $\cos(\arcsin\frac{3}{5})$ using a right-triangle diagram drawn on the board.

Active Monitoring

Circulate and check:

- Notes Section 2: Are students writing the correct interval notation (brackets vs. parentheses for tangent)?
- Notes Section 3: Are students identifying the *principal value* (angle in the restricted range), not just any angle?
- Cold-call prompts: "Why can't arcsin output $\frac{5\pi}{6}$?" and "What is the domain of arccos?"

Group Work & Differentiation

- **Tier R (Remediate):** Use a provided table matching trig values to angles; evaluate arcsin, arccos, arctan by look-up. Focus on reading the table correctly.
- **Tier A (Apply):** Determine domain and range of each inverse trig function; evaluate expressions like $\sin(\arccos\frac{3}{5})$ using right-triangle reasoning.
- **Tier E (Extend):** Compose trig and inverse trig functions analytically; determine precisely when $\sin(\arcsin x) = x$ vs. $\arcsin(\sin x) = x$ and explain in terms of restricted domains.

Individual Work & Assessment

Exit Ticket (2 items, 1 page):

1. Evaluate two inverse trig expressions (principal values).
2. Explain in 1–2 sentences why domain restriction is necessary for defining inverse trig functions.

Assessment note: Look for correct identification of the principal value and clear reasoning about one-to-one behavior.

Reinforcement & Extension

- **Homework:** 5–6 evaluation/domain-range problems, 2 AP-style MC items, 1 composition extension problem.
- **Extension:** Prove that $\arcsin x + \arccos x = \frac{\pi}{2}$ for all $x \in [-1, 1]$.
- **Preview Lesson 3.10:** Ask students to think about what it means to *solve* $\sin x = \frac{1}{2}$ — how many solutions are there? We will use inverse trig as a tool for solving trig equations.